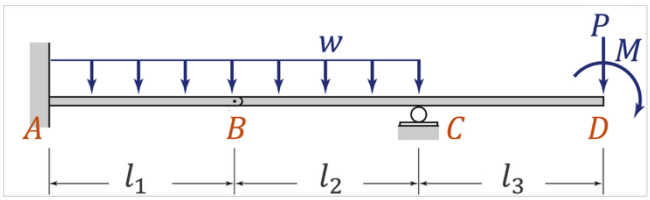
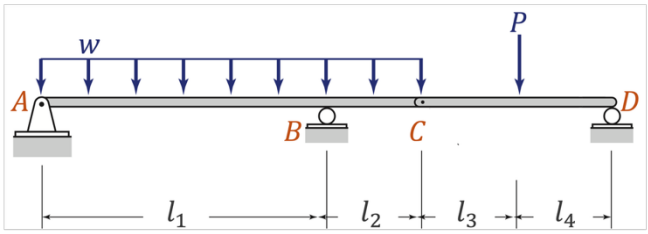
Quiz 05. Analysis of frames and machines
prepared by C. Cahig *for* ENS161 (AY 2026-2027, S1)

Instructions. Unless otherwise stated, express support reactions (forces and moments) and hinge forces as Cartesian vectors.



Scenario 1. The left part of the compound beam is fixed to the wall at A and is hinged to the right part at B. Let $l_1 = 14$ ft, $l_2 = 15$ ft, $l_3 = 16$ ft, $w = 2.3$ k/ft, $P = 5$ k, and $M = 30$ k-ft. Determine:

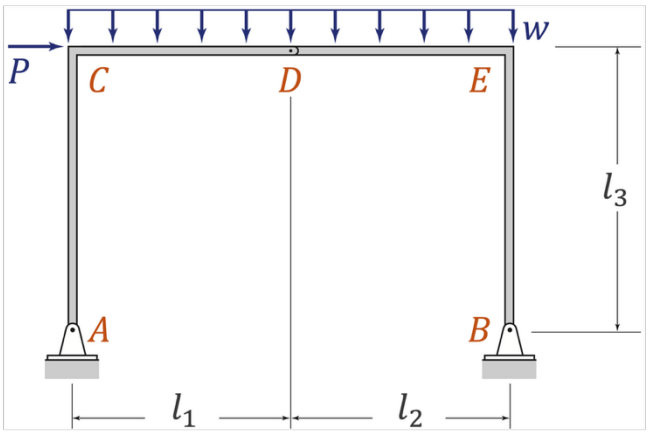
a. the reaction force at A,
b. the reaction moment at A,
c. the hinge force at B, and
d. the reaction at C.



Scenario 2. The two beams are connected by a pin at C. Determine:

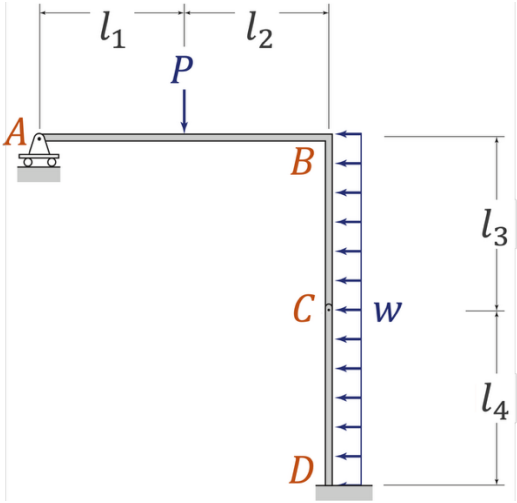
a. the reaction at A,
b. the reaction at B,
c. the hinge force at C, and
d. the reaction at D

when $l_1 = 30$ ft, $l_2 = 9$, $l_3 = 10$ ft, $l_4 = 11$ ft, $w = 1.0$ k/ft, and $P = 12$ k.



Scenario 3. The two angled members ACD and BED are joined by a hinge at D. For $P = 30$ k, $w = 2$ k/ft, $l_1 = 13$, $l_2 = 17$, and $l_3 = 21$, determine:

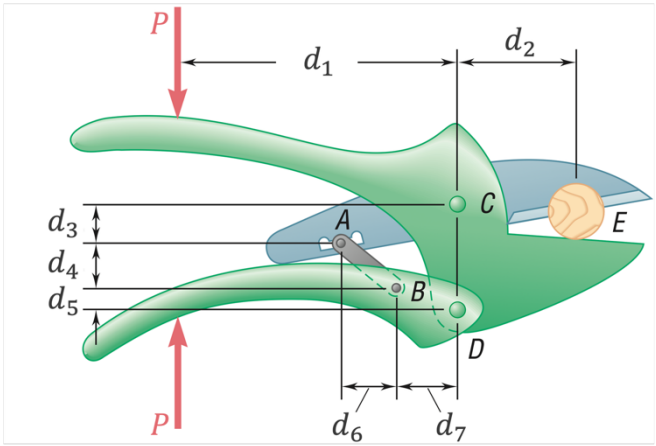
a. the reaction at A,
b. the reaction at B, and
c. the hinge force at D.



Scenario 4. Members ABC and CD are connected by a hinge at C. The bottom end of CD is embedded to a rigid floor.

a. the reaction at A,
b. the hinge force at C,
c. the reaction force at D, and
d. the reaction moment at D

when $P = 75$ kN, $w = 15$ kN/m, $l_1 = 6$ m, $l_2 = 7$ m, $l_3 = 6$ m, $l_4 = 9$ m.



Scenario 5. Consider the following dimensions: $d_1 = 3.5$ in, $d_2 = 1.6$ in, $d_3 = 0.5$ in, $d_4 = 0.55$ in, $d_5 = 0.25$ in, $d_6 = 0.65$ in, and $d_7 = 0.75$ in. Assume that the connections at A, B, C, and D are pins. If it takes 317-lb vertical forces to complete the pruning of the branch, determine:

a. the required value of P, and
b. the corresponding force in AB.

Marking. Each numbered item above is worth three (3) points. The total percentage rating a student gets for this Activity is computed as $0.59(0.9^m r) + 0.41p$, where m is the number of unmet requirements and violated instructions, r is the raw score (in percent), and p is the peer rating (in percent) by his/her groupmates.

Submission. Answers shall be communicated through the accompanying Google Form. Peer ratings shall be communicated via the submission platform in the Google Classroom.